

TrES-2b

R-1256

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-2_131107 · created 2026-07-31T12:15:34+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2456603.66276 +/- 0.00025 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.117 +/- 0.011
Transit depth (Rp/Rs)^2	1.38 +/- 0.27 [%]
Orbital Inclination (inc)	85.23 +/- 1.14
Airmass coefficient 1 (a1)	1.289 +/- 0.011
Airmass coefficient 2 (a2)	-0.1164 +/- 0.0026
Scatter in the residuals of the lightcurve fit is	0.87 %
Best Comparison Star	None
Optimal Aperture	3.97
Optimal Annulus	7.55
Transit Duration (day)	0.088 +/- 0.013

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.117 $\pm$ 0.011 vs 0.1254 (deviation 0.73 σ)
Duration fit vs published	0.088 d vs 0.0746 d (deviation 0.98 σ)
Mid-time O–C	0.5 min
Depth SNR	10.1
Residual scatter	0.87 %

Light curve

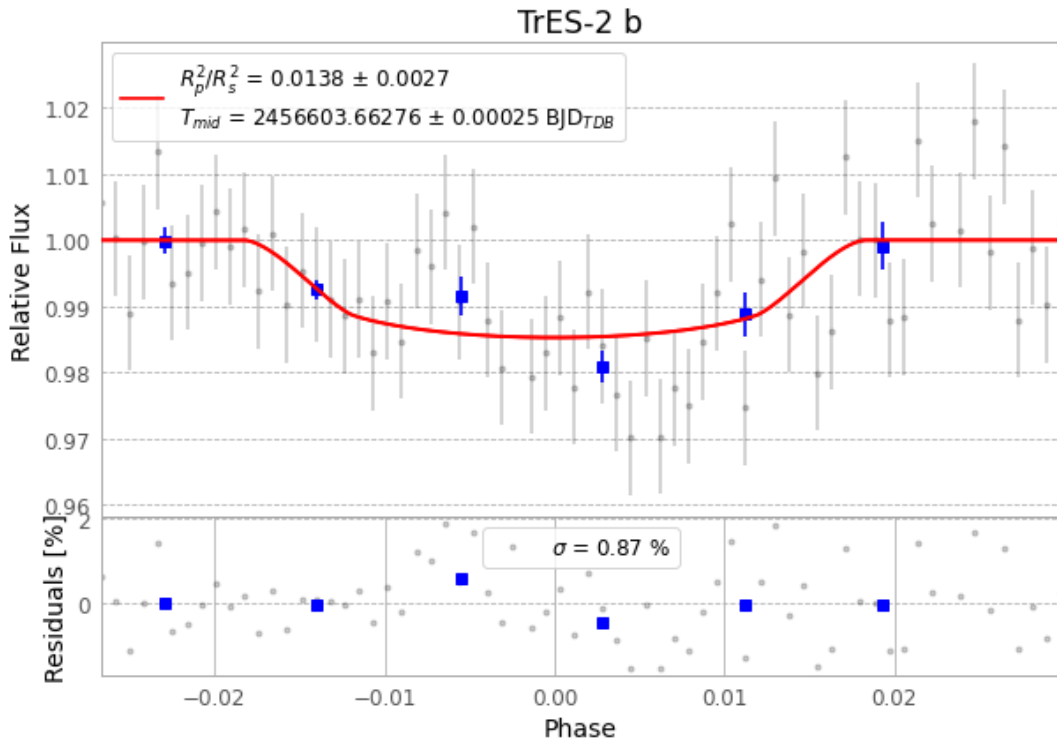


Figure 1 — FinalLightCurve_TrES-2 b_2013-11-06.png (SHA-256 7a76ca2080990ef6...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [306, 261], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 75a307935b32d628...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

68 FITS frames (45 MB), TRES-2131107021812.FITS → TRES-2131107053912.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-2-131107.log (49307215170f...), FinalLightCurve_TrES-2 b_2013-11-06.png (7a76ca208099...), FinalLightCurve_TrES-2 b_2013-11-06.csv (4edd549fec0e...)

Compute resources

Wall time 115.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 12:15Z.